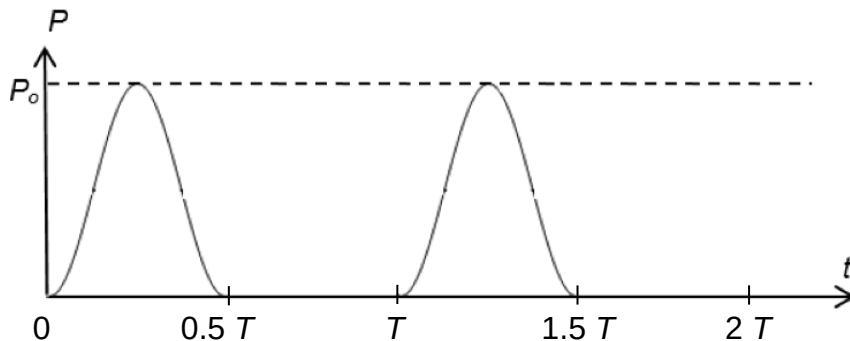


- 27 A half-wave rectified sinusoidal alternating current flows through a light bulb. The graph shows the variation of the power dissipated in the light bulb with time t , where T is the period of the current.



What is the average power consumption of the light bulb?

- | | | | | | | | |
|----------|------------|----------|------------|----------|------------|----------|------------|
| A | $0.25 P_0$ | B | $0.40 P_0$ | C | $0.50 P_0$ | D | $0.70 P_0$ |
|----------|------------|----------|------------|----------|------------|----------|------------|

Answer: A

Distribute area under graph across the entire period

For full sinusoidal a.c., the average power consumption is $0.50 P_0$.

But this is half-wave rectified sinusoidal a.c. the average power consumption is $0.5 (0.50 P_0) = 0.25 P_0$

- 28 To observe diffraction rings by a carbon film, a beam of electrons is accelerated from rest